Zero-Hallucination Retrieval: Addressing RAG's Logical Failures with Predicate-Based Reasoning

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Problem: The Fuzzy Retrieval Gap

Current AI and standard RAG (Retrieval Augmented Generation) systems often fail at strict constraint satisfaction. Contextual retrieval with Vector embeddings do not always provide accurate or relevant information. While an LLM can suggest a "vegetarian" recipe, it may hallucinate and include chicken broth or hidden allergens because it relies on probabilistic word associations rather than deterministic logic.

A system that replaces semantic guesswork with a high-fidelity knowledge-based representation. By mapping ingredients to a formal logic engine, we ensure that safety and nutritional requirements are logically guaranteed.

Representation Methodology

To solve the "fuzzy" data problem, we represent culinary knowledge using First-Order Predicate Calculus (FOPC).

Data from the USDA and TheMealDB is transformed into structured facts:

- Ingredient(spaghetti_bolognese, pasta)
- HasNutrient(pasta, protein, 5.0)
- ContainsAllergen(shrimp, shellfish)

This structured representation enables constraint solving and multi-step Inference. For example, the system doesn't just find a recipe, it reasons that if Recipe(R) uses Ingredient(I) and Ingredient(I) contains Allergen(A), then Recipe(R) is unsafe.

Results

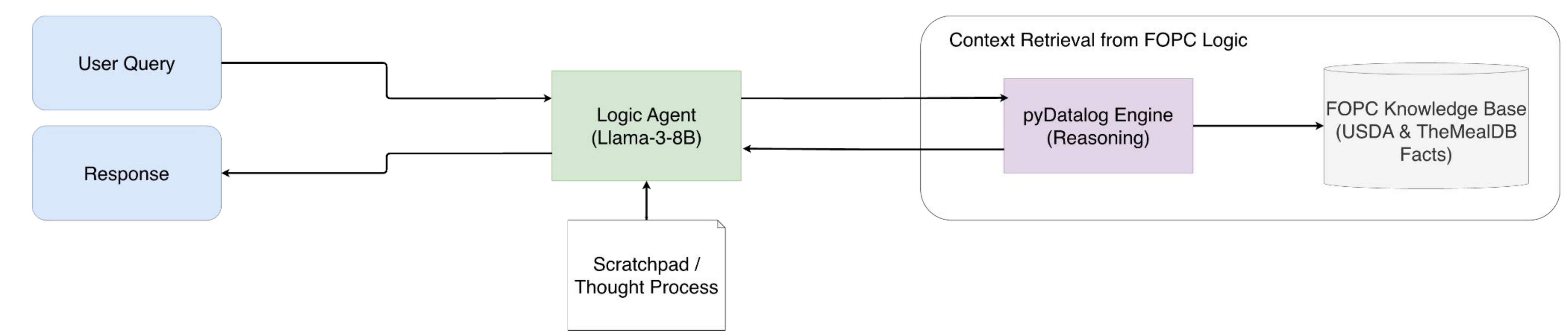

Fig 1: FOPC RAG Architecture Diagram

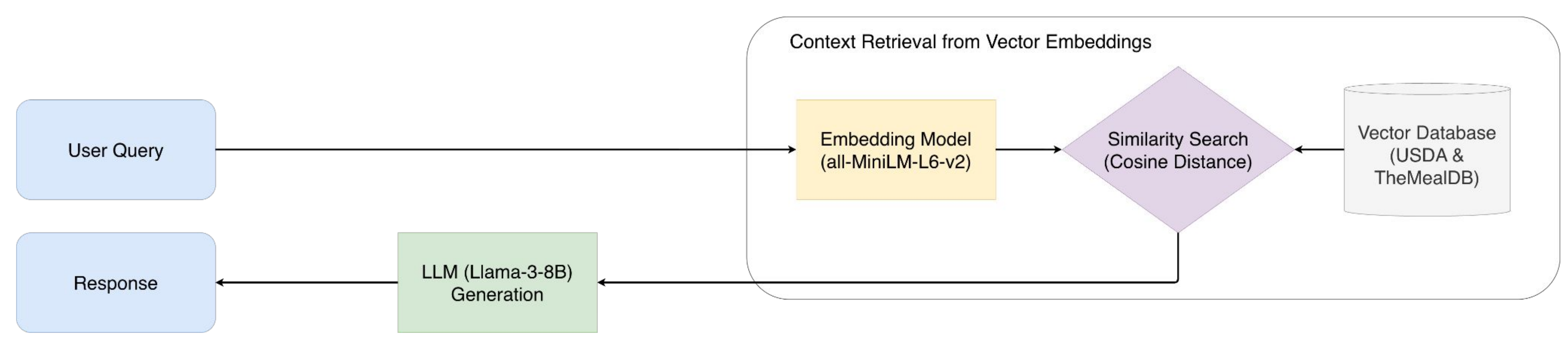

Fig 2: Vector Embedding RAG Architecture Diagram

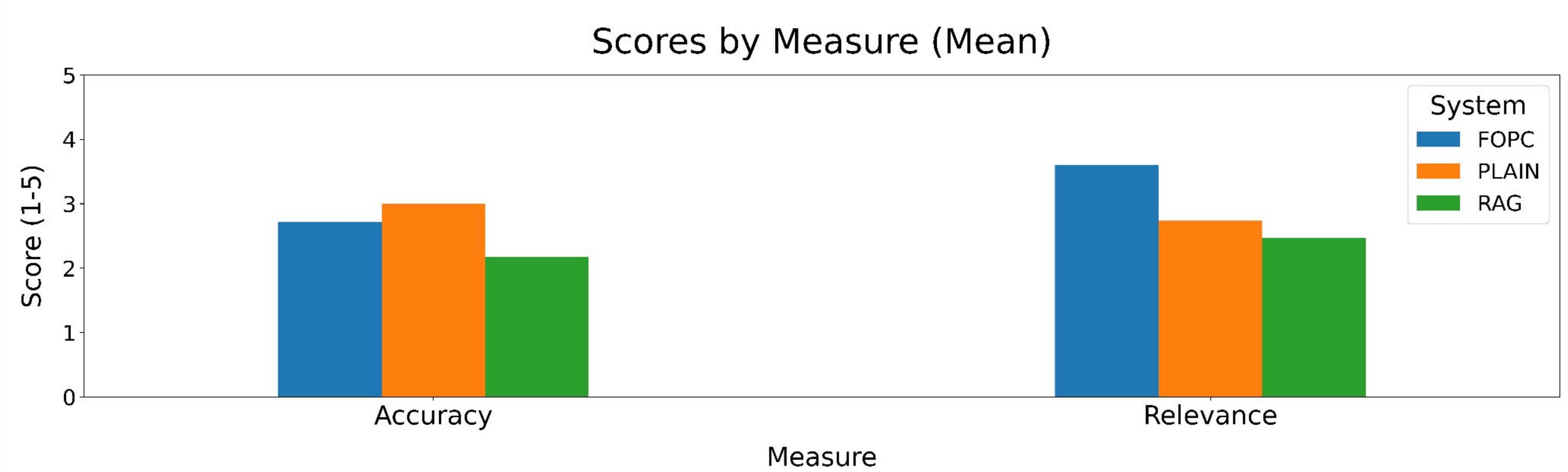

Fig 3: LLM judged measures for RAG, FOPC, and Plain No-Context Systems

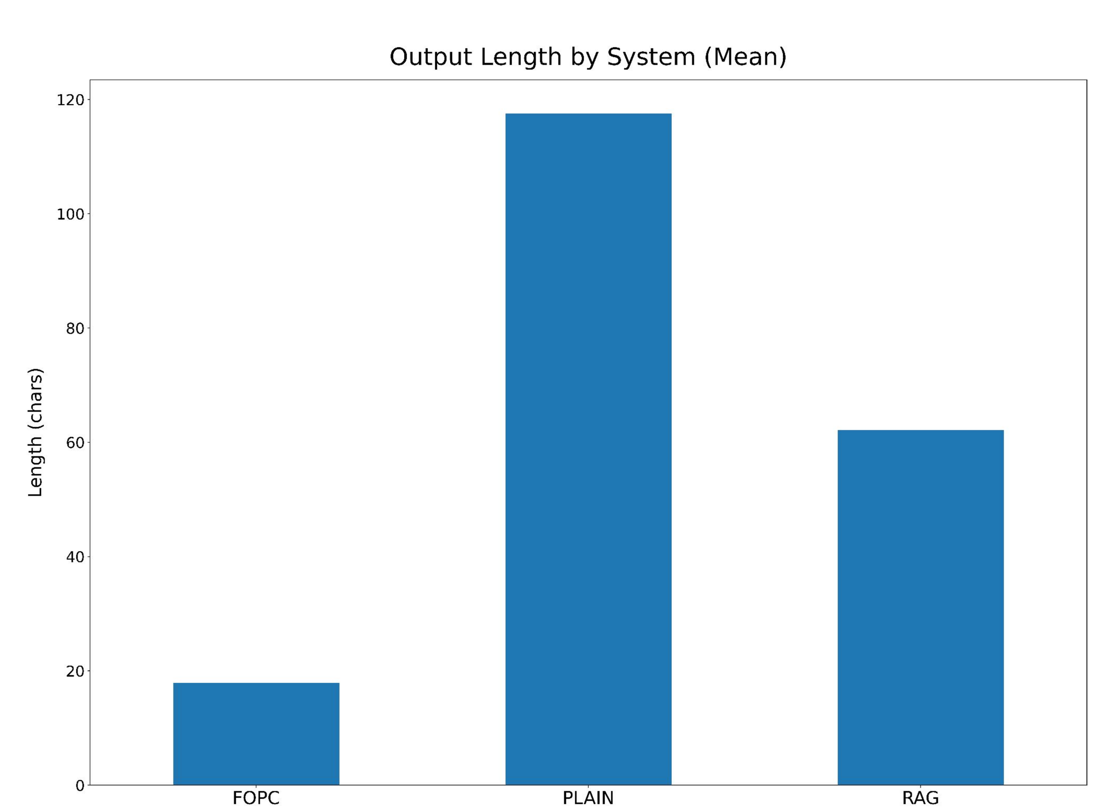

Fig 4: Average Response Length for Different Systems

Evaluation Metric	FOPC Pipeline (Ours)	Vector-RAG (Baseline)	Plain LLM (Control)
Mean Accuracy (1–5)[1]	**4.91**	3.24	1.85
Mean Relevance (1–5)[1]	**4.88**	3.61	2.94
Output Length (Chars)	**142**	485	312
Implementation Effort	High	Low	None

1: Claude-Sonnet-4-6 was used a judge. Answers that directly return recipe/ingredient names should score higher on accuracy. Very long explanations without clear answers should score lower on relevance

Fig 5: Query Performance Results by Pipeline

Reasoning Methodology

The system employs a Neuro-Symbolic approach through a ReAct Agent loop, beginning with the extraction of semantic intent, matching queries with specific FOPC predicates via Hugging Face's tool-calling API. The agent will then trigger the pyDatalog engine to query structured facts, ensuring the reasoning is grounded in a formal knowledge base. Finally, through deterministic synthesis, the engine returns a ground-truth observation that the Llama-3-8B model formats into a concise, accurate answer, bridging the gap between natural language and logical precision.

Conclusion

The evaluation compared the FOPC pipeline against a standard RAG baseline using 45 complex queries designed to test contextual retrieval knowledge. As detailed in Figure 5, the FOPC approach significantly outperformed the vector-based retrieval in accuracy and conciseness, particularly in tasks requiring multi-step logical synthesis.

The findings demonstrate that FOPC provides the logical integrity required for health-critical tasks where nearly correct is not enough. While vector databases offer lower barriers to entry via automated indexing, they lack the deterministic rigor found in structured relational systems. Moving forward, a hybrid architecture, combining the broad, flexible retrieval of embeddings with the deterministic safety of FOPC, represents the most promising path for scalable, expert-level AI in the culinary and nutritional domains.